

Predicting Certificate of Entitlement Price in Singapore

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1. Introduction

Car prices in Singapore are among the highest in the world (Julius Baer Group Ltd., 2025). A basic car can easily cost upwards of 150,000 Singapore dollars (S\$), which is around £88,000. More than half of a car's cost comes from the Certificate of Entitlement (COE), a vehicle quota system. As the name suggests, car buyers in Singapore must first obtain a COE in order to own a car.

COEs are distributed through an open-bidding auction held twice a month. Bidders submit their maximum willingness to pay, and certificates are allocated from the highest bidder downward. In the end, everyone pays the same price, called the Quota Premium (QP), which is the lowest successful bid plus S\$1. COEs are valid for only 10 years, after which owners can choose to renew them at the prevailing quota premium (a 3-month rolling average of QP) or scrap the car.

Different COEs are issued across vehicle categories. This paper focuses on Category A (Cat A) cars, which represent mainstream household vehicles. It is worth noting that Cat A prices may also be affected by Category B (Cat B) prices, which covers bigger, more powerful cars.

Figure 1 shows the Cat A COE price, which in recent years has crossed the S\$100,000 mark.

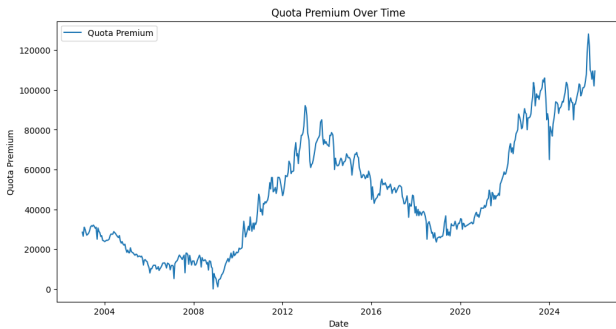


Figure 1. Cat A Certificate of Entitlement (COE) price over time.

In practice, car dealers bid on behalf of buyers, and they offer packages with different levels of guarantee for when a COE will be secured (Tan, 2013). Therefore, firms have an incentive to bid tactically to increase profits.

The literature currently lacks a paper that conducts a thorough predictive evaluation of COE prices. Instead, most papers focus on identifying the supply and demand determinants of COE premiums rather than assessing predictive performance (e.g., Chu & Goh (1997); Meng et al. (2015)). Unsurprisingly, these papers show that quota supply, past

prices, demand, and policy changes can all influence prices. However, explanatory power does not necessarily imply meaningful predictability. This is well documented in exchange-rate and stock-price forecasting. Although these financial time series can be expressed as functions of current and future economic conditions, the literature has consistently found a random walk difficult to beat (Hewamalage et al., 2023; Rossi, 2013). The first objective of this paper is to formally test whether the same holds for COE price prediction. The closest study is Kwan & Guan (2025), which used a Long Short-Term Memory model and found that historical prices and quota have the largest feature importance. However, that study did not compare predictions against a benchmark model, because its aim was to identify drivers rather than forecastability. From a forecasting perspective, if historical prices matter a lot, a random walk is already a strong benchmark. The second objective is to examine whether more complex machine-learning (ML) models can generate higher profits for firms. Due to the auction structure, standard metrics such as Mean Absolute Error (MAE) do not directly indicate how improved predictions translate into profits. To address this, we construct a custom profit metric.

Section 2 describe the data and methodology, section 3 the results, and section 4 concludes.

2. Method

2.1. Data

Cat A COE prices from 2003 Jan to 2026 Jan are used for prediction. Prices are converted into growth rates (with respect with the previous bid) to ensure stationarity¹, which can potentially improve learning (Makridakis et al., 2018). Predicted QP are then reconstructed with predicted growth rates.

Standard Supply, demand, and policy features are all included. Additionally, some non-standard features are included too. They are: Transport Consumer Price Index (CPI), Singapore Overnight Rate Average (SORA), Straits Times Index as a proxy of economic health, Google Trends data as proxies of demand, and certain Cat B information. Growth rates of features are included when sensible. A detailed list of features and data sources is in Appendix A1.

In total, there are 548 observations² and 55 features. Each

¹Growth rates are winsorised at 0.5% and 99.5%, which affects the top and bottom 3 observations in 2008, when COE prices fell to S\$2.

²In 2020, bidding was paused for 3 months due to COVID-19 measures

feature and the target is lagged 12 times. Missing values, which comes from CPI growth rates, gtrends, and SORA rates from early years, are filled with zero.

2.2. Exploratory Data Analysis

Table 1 shows features with a Pearson correlation above 0.2 with QP growth rate, plus `quota_per_bid`, which is the quota announced months before the bid³. The bolded features are those available at the time of bidding, which highlights the difference between identifying influence and making predictions. Although some features correlate strongly with the target contemporaneously, those correlations are much weaker when using only past data that is actually available at prediction time.

`quota_per_bid` does not correlate strongly with the target, and the strongest feature is past growth. The only other notable feature is `cat_a_divide_cat_b_lag_1`, which indicates how overpriced Cat A COEs are relative to Cat B COEs.

Table 1. Selected feature correlation with QP growth rate

Feature	Correlation
<code>quota_premium_cat_b_growth_rate</code>	0.568
<code>overdemand_growth_rate</code>	0.462
<code>overdemand_cat_b_growth_rate</code>	0.310
<code>quota_premium_growth_rate_lag_1</code>	-0.262
<code>overdemand</code>	0.259
<code>cat_a_divide_cat_b_lag_1</code>	-0.216
<code>quota_per_bid</code>	0.006
<code>quota_per_bid_growth_rate</code>	-0.047

Note: Only the bolded features are actually features used for prediction. `overdemand` = (bids received)/(quota). Only lagged-1 features are used, further lags are not evaluated.

2.3. Training and Evaluation Workflow

The last 96 observations were kept as the test set, with the remainder used for training. Three-fold Time Series Cross-Validation (TSCV) was applied. Specifically, models were trained on earlier data and validated on later data, with the training set expanding in each fold. A one-step-ahead walk-forward forecasting procedure was used to generate test-set predictions. After each prediction, the newly observed value was added to the training data and the model was retrained for the next step (hyperparameters unchanged). This is important because prices have reached record highs in recent years, and the model needs this newer data during training.

All models are implemented using the Darts Python library (Herzen et al., 2022), which provides a unified interface

³The actual quota is known only on the day of bidding and may differ slightly.

for time-series forecasting. Baseline models include a Random Walk (RW) and auto-ARIMA (selected MA(2)). Standard ML models include Elastic Net, Random Forest (RF), Support Vector Regression (SVR), and Extreme Gradient Boosting (XGB). Typical hyperparameters are tuned via grid search with MAE as the metric. The full specification is provided in Appendix A2. Deep learning with only about 500 observations is unlikely to perform well, but transfer learning may be a practical workaround. Here, we fine-tune Chronos-2-small Ansari et al. (2025). Chronos-2 adapts an encoder architecture commonly used in natural language processing for time-series forecasting. Its context consists of non-overlapping patches of time-series data and their covariates. Embeddings are passed through time attention layers to learn dependencies over time, and through group attention layers to learn relationships among covariates. The model is pre-trained on a large collection of public and synthetic time-series data, including financial series (Ansari et al., 2025).

Next, we evaluate the profitability of these models. The setup is as follows: suppose a car’s cost consists only of the COE. A firm sells a car between times $t - 1$ and t with a guarantee that the COE will be secured within two periods. The firm sets the car price, $price_t$, equal to the model prediction at time t , plus a margin.

We further assume that, at time t , the firm bids exactly what the model predicts. If the firm wins, its profit is $price_t - QP_t$, which is positive. If the firm loses, it bids very high at $t + 1$ to guarantee a COE. The firm’s profit is then $price_t - QP_{t+1}$, which can be positive or negative.

We use a logistic function to simulate demand, as shown in equation (1), where consumers anchor their price expectations to the most recent QP. If the firm sells above that price, demand (or the probability of purchase) falls, with the steepness governed by k . Conversely, if the firm sells below that price, demand rises.

$$\text{Demand}_t = \frac{1}{1 + e^{k(\text{Price}_t - \text{QP}_{t-1})}} \quad (1)$$

We use the previously trained model and evaluate it using this profit metric. MAE remains a relevant training objective because firms have an incentive to price as close to the QP as possible. Pricing too far above the QP leads to lost customers, while pricing too far below it increases the risk of losses.

3. Results

Table 2 reports model performance on level targets. The best model is XGB. Visual inspection of the prediction plots is not informative, because the differences are too small relative to the scale of COE prices (Appendix A3). Therefore,

we use the Diebold-Mariano (DM) test (Diebold & Mariano, 2002) to evaluate differences in absolute loss. XGB performs significantly better than Random Walk (p-value = 0.02), but not Auto-ARIMA (p-value = 0.50). This suggests that traditional econometric baselines remain difficult to beat. Nonetheless, there is some evidence in favor of XGB, so we further evaluate model performance using the profit metric. We also train an additional XGB model, called XGB_custom, where hyperparameters are tuned using the custom metric under $k = 0.0005$ and margin = 2000. The DM-test p-values for XGB_custom versus RW and ARIMA are 0.07 and 0.57, respectively.

Table 2. Test-set forecasting performance (3 d.p.)

Model	RMSE (levels)	MAE (levels)
Random Walk	5209.96	3823.12
AutoARIMA	5415.17	3760.74
Elastic Net	5263.12	3680.18
XGB	5046.38	3622.02
RF	5123.35	3726.58
SVR	5230.20	3755.01
Chronos-2	5421.15	3967.85
XGB_Custom	5048.38	3646.77

Figure 2 shows profits over a reasonable range of k values (0.0001 to 0.0025) at a margin of 2000. We compare XGB models against Random Walk and ARIMA. Lower k penalizes over- and underprediction less severely than higher k ⁴. The sharp drop in ARIMA's profit indicates that its predictions are the most inconsistent. XGB_custom and XGB both outperform ARIMA and Random Walk. This result is also robust across different margin values (Appendix A3).

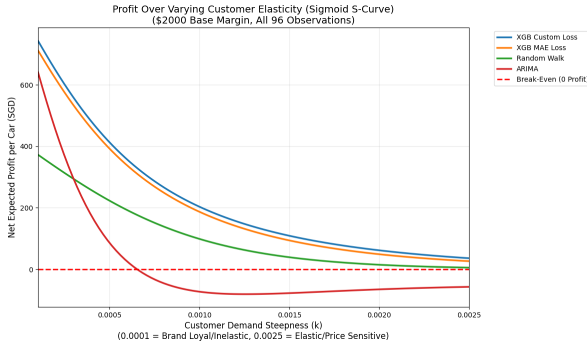

 Figure 2. Expected profit comparison across demand steepness k .

Table 3 reports the DM test on profit differentials at $k = 0.0005$ and a margin of 2000. Profit increases are substantial for both models relative to the stronger baseline, RW,

⁴A k of 0.0005 means demand falls from 0.50 to 0.38 given an S\$1000 increase in price relative to the expected price. This is already a large drop relative to COE price levels. Hence, model performance may be more realistically evaluated at smaller k values.

with XGB_custom performing better. This suggests that using the custom metric for hyperparameter tuning helps the model better balance the risk between underprediction and overprediction.

Table 3. Profit-differential comparison against Random Walk

Metric	XGB vs RW	XGB_custom vs RW
Profit difference	169.00 (76% increase)	189.56 (85% increase)
p-value	0.0062	0.0045

If firms want to understand what drives predictions, SHAP values are a useful tool for evaluating a single prediction. Table 4 reports feature importance for the final prediction in the test set. SHAP values for base features are calculated by summing the SHAP values of their respective lagged features. Absolute SHAP captures only magnitude, whereas SHAP values capture both magnitude and direction. The two features with the largest SHAP values correspond to those with the highest correlations in Table 1, followed by overdemand growth in Cat A and Cat B, and Cat A deregistration growth (which affects quota supply).

Table 4. SHAP values for the last test-set prediction

Base feature	SHAP value	Absolute SHAP
QP growth rate	0.57	1.21
cat_a_divide_cat_b	0.57	0.60
overdemand_cat_b_growth_rate	0.13	0.59
overdemand_cat_a_growth_rate	-0.18	0.37
deregis_cat_a_growth_rate	-0.23	0.28

Note: Lag 0 is not included as these are not available at time of prediction.

4. Conclusion

There is evidence that XGB can exploit covariate information to improve predictions. Although the improvement is on the order of hundreds, it still translates into more than 75% increase in expected profits under the assumed setup. The model may not fully reflect reality, but it illustrates the point that better QP predictions can translate into higher business profits.

Further research could train models on a classification task to predict whether COE prices will rise or fall next period. In addition, instead of using RW/ARIMA as a statistical baseline, one could compare XGB's performance with firms' existing performance.

Github repo in case anything goes wrong with github classroom submission: https://github.com/renjielim/d200_project

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A. Appendix

A.1. Features and Data Source

Table 3 displays the full feature lists, where there is 1 target variable and 55 features in total. All features aligned to ensure data is available at point of estimation. 'Gr.' means if growth rate of the variable is included, and 'B?' means if Cat B of the feature is included.

Table A.1: Feature list and data sources

Feature	Gr.	B?	Description	Source	Frequency
quota_premium	Yes	Yes	COE clearing price	SingStat	Fortnightly
quota	Yes	No	COE supply released pre-auction		Fortnightly
quota_per_bid	Yes	No	Approximate supply announced beforehand	LTA	Quarterly
overdemand	Yes	Yes	Bids received divided by quota	SingStat	Fortnightly
cat_a_divide_cat_b	No	No	Cat A QP over Cat B QP		Fortnightly
prevailing_quota_premium	Yes	Yes	3-month moving average of QP		Monthly
total_car_registered	Yes	Yes	Total car stock		Monthly
new_car_regis_growth_rate	NA	No	Growth in new registrations		Monthly
deregistrations	Yes	No	Vehicles removed from registry		Monthly
cpi_growth_rates	NA	NA	10 CPI sub-indexes		Monthly
sora_1m	No	NA	1-month Moving Average overnight benchmark rate	MAS	Fortnightly
straits_times_index	Yes	NA	Economic health proxy	Yahoo Finance	Fortnightly
google_trends	No	NA	Search trends: COE and sgcar mart	Google Trends	Monthly
policy_dummies	NA	NA	17 dummies for 10 policy win-dows	The Straits Times	NA

A.2. Hyperparameters

Table A2 summarises the main hyperparameter grids used for model tuning. 12 lags are always used. 3 fold tscv is used for tuning, with MAE as metric for all models except XGB_custom. Where possible, the full training is also based on MAE.

Note Chornos is not tuned due to computational limitations.

Table A.2: Grid-search hyperparameters by model

Model	Hyperparameter	Grid	Actual parameter selected
Elastic Net	alpha	{.01, 1, 2, 5, 10, 20, 30}	30
	l1_ratio	{0, .2, .4, .6, .8, 1.0}	1
Random Forest	n_estimators	{800}	800
	max_depth	{3, 5, 8}	5
	min_samples_split	{5, 10}	5
	min_samples_leaf	{2, 5}	5
	max_features	{Square Root}	Square Root
SVR	kernel	{rbf, linear}	rbf
	C	{.3, 1.0, 3.0, 10.0, 30.0}	1.0
	epsilon	{.1, .5, 1, 2}	2
	gamma	{scale}	scale
XGB	n_estimators	{200, 500}	200
	learning_rate	{.01, .05}	.01
	max_depth	{3, 5}	5
	subsample	{.6, .8, 1.0}	.8
	colsample_bytree	{.8, 1.0}	.8
	reg_alpha	{.1, .3, .5}	.1

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Model	Hyperparameter	Grid	Actual parameter selected
XGB_custom	n_estimators	{200, 500}	200
	learning_rate	{0.01, 0.05}	.01
	max_depth	{3, 5}	5
	subsample	{.6, .8, 1.0}	1.0
	colsample_bytree	{.8, 1.0}	.8
	reg_alpha	{.1, .3, .5}	.3
Chronos-2	model	autogluon/chronos-2-small	—
	batch_size	{32}	—
	lr	{1e-3}	—
	n_epochs	{3}	—
	gradient_clip_val	{1}	—
	Optimiser	AdamW (default weight decay of 0.01)	—

A.3. results

Figure A1 shows the time series plot of the XGB and RW predictions, against the actual. Figure A2 shows the different profit curve for different margin levels.

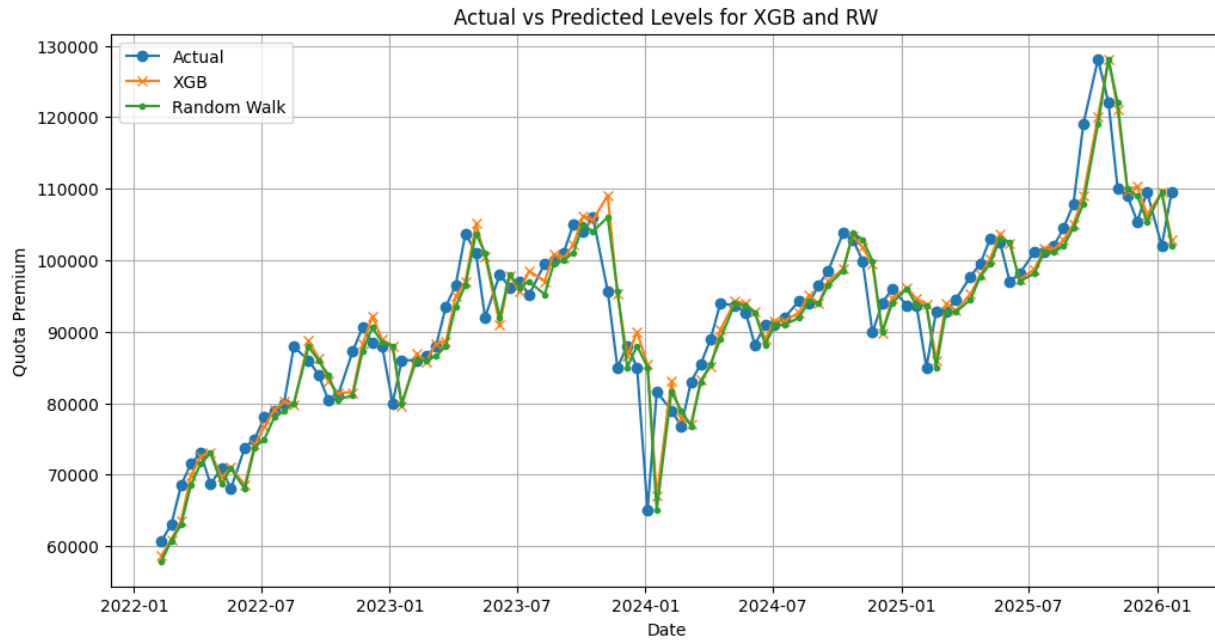


Figure A.1. Actual vs XGB vs Random Walk on test set. XGB and Random Walk is almost indistinguishable

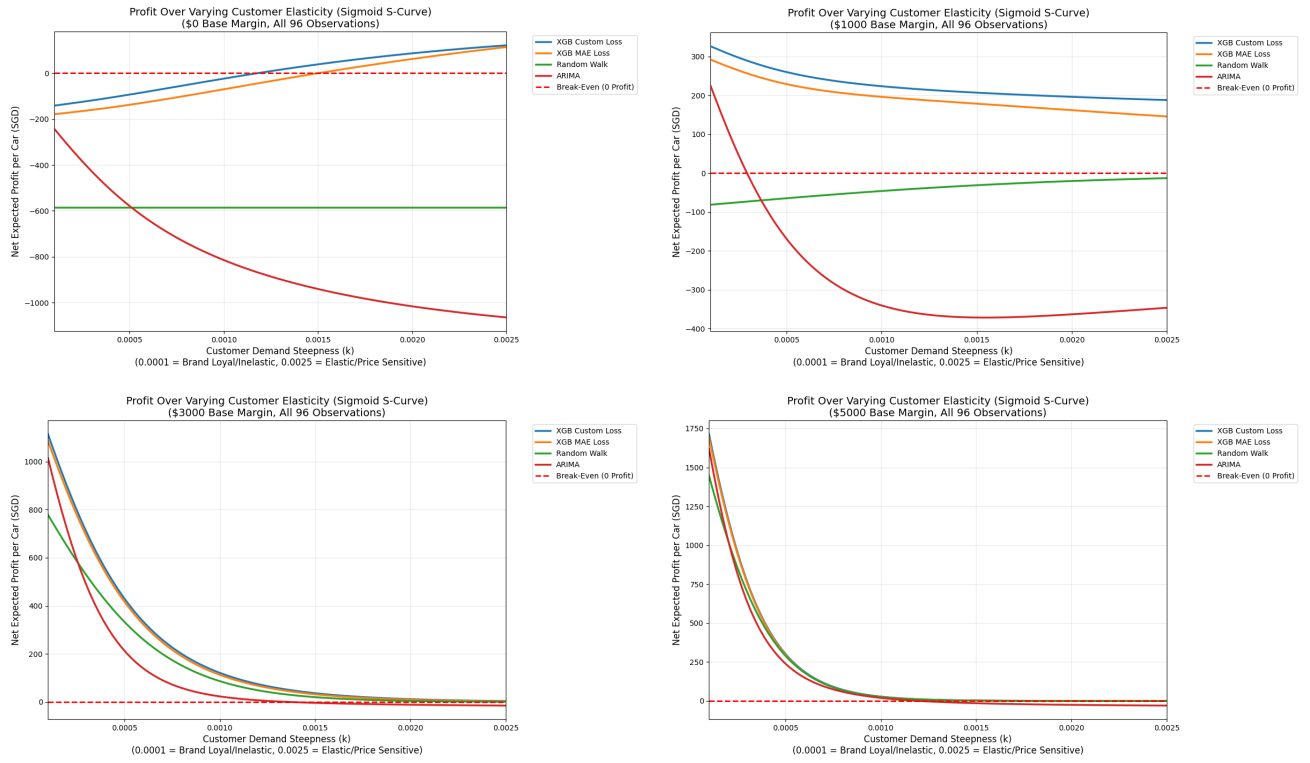


Figure A.2. Profit across different margins. The result that XGB does better is robust across different margin levels. The difference shrinks at high margin levels because the profits are driven by margins rather than predictions.